

CHAPTER 2:

Chapter 1: Problem 1, 2, 10, 13 and 14

Chapter 2: Problem 2, 11, 13

Problem 2/p.75 Lottery winner.

SOLVE

- We have: €0.5988/AUD

- The number of AUDs can Aisha get = $€150 \times \frac{1}{0.5988} \text{AUD/€} = \text{AUD}250.50$

- The value date of the AUD payment ?

Problem 11/p.76 Vietnamese Coffee Coyote.

SOLVE

- Vietnamese bank rate: After exchange, \$1 = 19,800 - 19,800 x 2.5% = 19,305VND

- Saigon Airport exchange bureau rate = After exchange, \$1 = 19,500 - 19,500 x 2.0% = 19,012.5VND

- Hotel exchange bureau rate: After exchange, \$1 = 19,400 - 19,400 x 1.5% = 19,109VND

=> Vietnamese bank rate would give the best interest.

CHAPTER 5: The Foreign Exchange Market

Chapter 5: The Foreign Exchange Market (Problem 2, 4, 7, 8 and 14, 15, 16)

Problem 2: Victoria Exports.

Period	Days Forward	Canadian\$/Euro	US\$/Euro
Spot		1.3360	1.3221
1 month	30	1.3368	1.3230
2 months	60	1.3376	1.3228
3 months	90	1.3382	1.3224
6 months	180	1.3406	1.3215
12 months	360	1.3462	1.3194

SOLVE

• C\$/Euro:

Period	Days Forward	C\$/Euro	Forward Premium on the Canadian\$/Euro (on Euro)	Canadian\$ Proceeds of €12,000	Difference Over Spot
Spot		1.3360		= €12,000 x C\$1.3360/€ = C\$16,032	
1 month	30	1.3368	= (1.3368 - 1.3360)/1.3360 x 100 x (360/30) = 0.7186%	= €12,000 x C\$1.3368/€ = C\$16,041.6	= 16,041.6 - 16,032 = C\$9.6
2 months	60	1.3376	= (1.3376 - 1.3360)/1.3360 x 100 x (360/60) = 0.7186%	= €12,000 x C\$1.3376/€ = C\$16,051.2	= 16,051.2 - 16,032 = C\$19.2
3 months	90	1.3382	= (1.3382 - 1.3360)/1.3360 x	= €12,000 x C\$1.3382/€ = C\$16,058.4	= 16,058.4 - 16,032

6 months	180	1.3406	$100 \times (360/90) = 0.6587\%$ $= (1.3406 - 1.3360)/1.3360 \times 100 \times (360/180) = 0.6886\%$ $= (1.3462 - 1.3360)/1.3360 \times 100 \times (360/360) = 0.7635\%$	$= €12,000 \times C\$1.3406/€ = C\$16,087.2$ $= €12,000 \times C\$1.3462/€ = C\$16,154.4$	$= C\$26.4$ $= 16,087.2 - 16,032 = C\$55.2$ $= 16,154.4 - 16,032 = C\$122.4$
12 months	360	1.3462			

• US\$/Euro:

Period	Days Forward	US\$/Euro	Forward Premium on the Canadian\$/Euro (on Euro)	US\$ Proceeds of €12,000	Difference Over Spot
Spot		1.3221			
1 month	30	1.3230	$= (1.3230 - 1.3221)/1.3221 \times 100 \times (360/30) = 0.817\%$	$= €12,000 \times \$1.3221/€ = \$15,865.2$ $= €12,000 \times \$1.3230/€ = \$15,876$	$= 15,876 - 15,865.2 = \$10.80$
2 months	60	1.3228	$= (1.3228 - 1.3221)/1.3221 \times 100 \times (360/60) = 0.318\%$	$= €12,000 \times \$1.3228/€ = \$15,873.6$	$= 15,873.6 - 15,865.2 = \8.40
3 months	90	1.3224	$= (1.3224 - 1.3221)/1.3221 \times 100 \times (360/90) = 0.091\%$	$= €12,000 \times \$1.3224/€ = \$15,868.8$	$= 15,868.8 - 15,865.2 = \3.60
6 months	180	1.3215	$= (1.3215 - 1.3221)/1.3221 \times 100 \times (360/180) = -0.091\%$	$= €12,000 \times \$1.3215/€ = \$15,858$	$= 15,858 - 15,865.2 = -\$7.2$
12 months	360	1.3194	$= (1.3194 - 1.3221)/1.3221 \times 100 \times (360/360) = -0.204\%$	$= €12,000 \times \$1.3194/€ = \$15,832.8$	$= 15,832.8 - 15,865.2 = -\32.4

=> **CONCLUDE:** Since the company wishes to lock in the forward rate for each and every payment, it would appear that the company should lock in forward rates in C\$ for all payments. Since the euro is selling forward at a greater premium against the Canadian dollar than the U.S. dollar, the resulting dollar proceeds are higher.

Problem 4: Credit Suisse Geneva.

SOLVE

a. Calculate outright quotes for bid and ask, and the number of points spread between each

	Bid	Ask	Spread
One-month forward	$1.2575 + 0.001 = 1.2585$	$1.2585 + 0.0015 = 1.2600$	$1.2600 - 1.2585 = 0.0015$
3-month forward	$1.2575 + 0.0014 = 1.2589$	$1.2585 + 0.0022 = 1.2607$	$1.2607 - 1.2589 = 0.0018$
6-month forward	$1.2575 + 0.002 = 1.2595$	$1.2585 + 0.003 = 1.2615$	$1.2615 - 1.2595 = 0.0020$

b. What do you notice about the spread as quotes evolve from spot forward six months?

- It widens, most likely a result of thinner and thinner trading volume.

c. What is the 6-month Swiss bill rate?

- The Swiss bill rate for 6-months:

$$= \text{Six-month US dollar treasury rate} + \left[\left(\frac{\text{Six month forward rate}}{\text{Spot rate}} \right)^{1/(\text{maturity} / 360)} - 1 \right]$$

$$= 4.2\% + \left[\left(\frac{\text{Six month forward rate}}{\text{Spot rate}} \right)^{1/(\text{maturity} / 360)} - 1 \right]$$

Problem 7: Asian Pacific Crisis.

SOLVE

a. Which of the following currencies had the largest depreciation or devaluations during the July to November period?

Country	Currency	July 1997 (per US\$)	November 1997 (per US\$)	Percentage Change vs dollar
China	yuan	8.40	8.40	$= (8.40 - 8.40)/8.40 = 0\%$
Hong Kong	dollar	7.75	7.73	$= (7.75 - 7.73)/7.73 = 0.26\%$
Indonesia	rupiah	2,400	3,600	$= (2,400 - 3,600)/3,600 = -33.3\%$
Korea	won	900	1,100	$= (900 - 1,100)/1,100 = -18.18\%$
Malaysia	ringgit	2.50	3.50	$= (2.50 - 3.50)/3.50 = -28.57\%$
Philippines	peso	27	34	$= (27 - 34)/34 = -20.59\%$
Singapore	dollar	1.43	1.60	$= (1.43 - 1.60)/1.60 = -10.62\%$
Taiwan	dollar	27.80	32.70	$= (27.80 - 32.70)/32.70 = -14.98\%$
Thailand	baht	25.0	40.0	$= (25 - 40)/40 = -37.50\%$

b. Which seemingly survived the first 5 months of the crisis with the least impact on their currencies ?

- The Chinese yuan's value against the US dollar, as a result of the Chinese government maintaining its peg to the dollar, did not change at all during the crisis. The Thai baht, however, fell 37.5% in only 5 months, with the Indonesian rupiah a close second with a loss of 33.3%.

Problem 8: Bloomberg Currency Cross Rates.

SOLVE

- USD = 83.735 JPY
- JPY = 0.0119 USD = 1/83.735
- EUR = 1.3247 USD
- USD = 0.7549 EUR = 1/1.3247
- EUR = 110.9238 JPY

- f. JPY = 0.009 EUR = 1/110.9238
- g. USD = 1.0097 CAD
- h. CAD = 0.9904 USD = 1/1.0097
- i. USD = 1.015 AUD
- j. AUD = 0.9852 USD = 1/1.015
- k. USD = 0.6328 GBP
- l. GBP = 1.5804 USD = 1/0.6328
- m. CHF = 1.0184 USD
- n. USD = 0.9819 CHF = 1/1.0184

Problem 14: Venezuelan Bolivar (B). The Venezuelan political and economic crisis deepened in late 2002 and early 2003. On January 1, 2003, the bolivar was trading at Bs1400/\$. By February 1, its value had fallen to Bs1950/\$. Many currency analysts and forecasters were predicting that the bolivar would fall an additional 40% from its February 1 value by early summer 2003.

SOLVE

Beginning	Bs1400/\$
Ending	Bs1950/\$

a. What was the percentage change in January?

Bs is price currency, \$ is unit currency

$$\Rightarrow \text{Percentage change (\% } \Delta) = \frac{\text{Beginning rate} - \text{Ending rate}}{\text{Ending rate}} \times 100 = \frac{1400 - 1950}{1950} \times 100 = -28.205$$

b. What is the forecast value for June 2003?

Bs is price currency, \$ is unit currency

- Beginning - February 01: Bs1,950/\$
- Bs fall 40% $\Rightarrow$ % change = -40%

$$\Rightarrow \% \text{ change} = \frac{\text{Bs } 1950/\$ - \text{Ending}}{\text{Ending}} \times 100 = -40 \Rightarrow \text{Ending} = \text{Bs}3,250/\$$$

$\Rightarrow$ The forecast value for June 2003 is Bs3,250/\$

Problem 15: Indirect on the Dollar. Calculate the forward premium on the dollar (the dollar is the home currency) if the spot rate is €1.3300/\$ and the 3-month forward rate is €1.3400/\$.

SOLVE

- Spot rate (S) = €1.3300/\$
- Forward rate (F) = €1.3400/\$
- 3-month = 90 days
- Dollar is unit currency, Euro is price currency

$\Rightarrow$ To calculate forward premium on USD, we use the formula: $((F - S)/S) \times 100 \times 360/\text{number of days}$

$$\Rightarrow \text{Forward premium on USD} = ((1.34 - 1.33)/1.33) \times 100 \times 360/90 = 3\%$$

$\rightarrow$ SAI

Problem 16: Direct on the Dollar. Calculate the forward discount on the dollar (the dollar is the home currency) if the spot rate is \$1.5800/£ and the 6-month forward rate is \$1.5550/£.

SOLVE

- Spot rate (S) = \$1.5800/£
- Forward rate (F) = \$1.5550/£
- 6-month = 180 days
- Dollar is price currency, Euro is unit currency

=> To calculate forward premium on USD, we use the formula: $((S - F)/F \times 100 \times 360/\text{number of days})$

=> Forward premium on USD = $((1.58 - 1.555)/1.555 \times 100 \times 360/180 = 3.215\%$

-> SAI

CHAPTER 6: International Parity Condition

Problem 1: Pulau Penang Island Resort.

SOLVE

Assumptions:

- Charge for suite plus meals in Malaysia ringgit (RM) = RM1,000/day
- Spot exchange rate (MYR/GBP) = RM5.2522/GBP => GBP0.19034/RM
- GBP cost today for a 30-day stay = $1,000 \times 1/5.2522 \times 30 \text{ days} = \text{GBP}5,711.89$

a. How many pounds might Theresa expect to need one year hence to pay for her 30-day vacation?

- Spot exchange rate (MYR per GBP) = RM5.2522/GBP
- Malaysian ringgit inflation rate expected to be = 3%
- GBP inflation rate expected to be = 1%

$\text{Spot (expected in 1 year)} = \text{Spot} \times (1 + \text{RM inflation}) / (1 + \text{GBP inflation})$
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=> Expected spot rate one year from now based on PPP (RM/GBP) = $5.2522 \times (1 + 3\%) / (1 + 1\%) = \text{RM}5.356204/\text{GBP}$

=> Expected hotel charges to be paid one year from now for a 30-day stay in MYR = $\text{RM}1,000 \times (1 + 3\%) \times 30 = \text{RM}30,900$

- GBP needed on the basis of these two expectations = $\text{RM}30,900 \times \frac{1}{\text{RM}5.356204/\text{GBP}} = \text{RM}30,900 = \text{GBP}5,769.01$

b. By what percent will the pound cost have gone up ? Why ?

New dollar cost	GBP5,769.01
Original dollar cost	GBP5,711.89
=> Percent change in GBP cost = $(5,769.01 - 5,711.89)/5,711.89 = 0.01 = 1\%$	

=> The GBP cost has risen by the UK inflation rate. This is a result of your estimation of the future GBP costs and the exchange rate changing in proportion to inflation (relative purchasing power parity).

Problem 2: Argentine Tears.

SOLVE

Assumptions:

- Spot exchange rate, fixed peg, early January 2002 (Ps/\$) = Ps1.00/\$
- Spot exchange rate, January 29, 2003 (Ps/\$) = Ps3.20/\$
- US inflation for year (per annum) = 2.2%
- Argentine inflation for year (per annum) = 20%

a. What should have been the exchange rate in January 2003 if PPP held?

- Beginning spot rate = Ps1.00/\$
- Argentine inflation = 20%
- US inflation = 2.20%

=> PPP spot exchange rate = $1.00 \times (1 + 20\%) / (1 + 2.2\%) = \text{Ps}1.1742/\$$

b. By what percentage was the Argentine peso undervalued on an annualized basis?

- Actual exchange rate = Ps3.20/\$
- PPP exchange rate = Ps1.17/\$

=> Percentage overvaluation (positive) or undervaluation (negative) = $\frac{1.17 - 3.20}{3.20} = -0.6344 = -63.44\%$

c. What were the probable causes of undervaluation ?

- The rapid decline in the value of the Argentine peso was a result of not only inflation but also a severe crisis in the balance of payments.

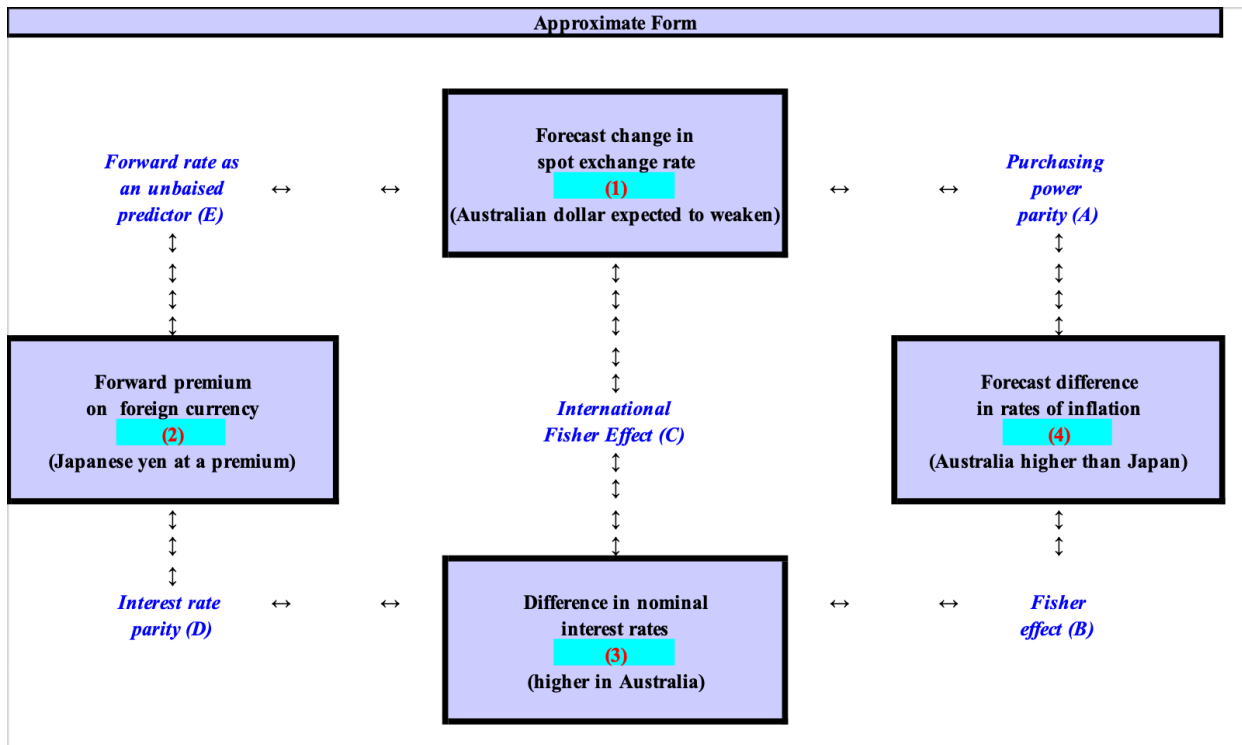
Problem 3: Japanese/United States Parity Conditions.

SOLVE

Assumptions:

- Forecast annual rate of inflation for Japan = 5%
- Forecast annual rate of inflation for Australia = 6.8%
- One-year interest rate for Japan = 7.85%
- One-year interest rate for Australia = 9.7%
- Spot exchange rate = ¥108.33/A\$
- One-year forward exchange rate = ¥106.50/A\$

a. Diagram and calculate whether international parity conditions hold between Japan and Australia?



+ (1): Forecast change in spot exchange rate (Australian dollar expected to weaken):

$$= \frac{108.33 - 106.50}{106.50} = 1.718\%$$

+ (2): Forward premium on foreign currency (Japanese yen at a premium):

$$= \frac{108.33 - 106.50}{106.50} = 1.718\%$$

+ (3): Difference in nominal interest rates (higher in Australia) = 7.850% - 9.7% = -1.850%

+ (4): Forecast difference in rates of inflation (Australia higher than Japan) = 5% - 6.8% = -1.8%

=> As is always the case with parity conditions, the future spot rate is implicitly forecast to be equal to the forward rate, the implied rate from the international Fisher effect, and the rate implied by purchasing power parity. According to the calculations, the markets are indeed in equilibrium - parity - and the future spot rate is projected to be JPY106.50/AUD.

b. Find the forecasted change in the Japanese yen/Australian dollar (¥/A\$) exchange rate one year from now.

- Spot exchange rate = ¥108.33/A\$

- One-year forward exchange rate = ¥106.50/A\$

$$\begin{aligned} \Rightarrow \text{Forecasted change in exchange rates} &= \frac{\text{Current Spot Rate} - \text{Forward Exchange Rate}}{\text{Forward Exchange rate}} \\ &= \frac{108.33 - 106.50}{106.50} = 1.7183\% \end{aligned}$$

Problem 4: Sydney to Phoenix. (DONE)

SOLVE

Assumptions:

- Price of 3-Piece Luggage set in US\$ = US\$850
- Price of 3-Piece Luggage set in A\$ = A\$930
- Spot exchange rate = A\$1.0941/\$
- US inflation for year (per annum) = 1.15%
- Australian inflation for year (per annum) = 3.13%

a. Is the price of the luggage truly equal if the spot rate is A\$1.0941/\$?

- Spot exchange rate as determined by PPP = Price in A\$ / Price in US\$ = 930 / 850 = A\$1.0941/US\$

b. What should be the price of the luggage set in A\$ in 1-year if PPP holds? ?

- Beginning spot rate (A\$/) = A\$1.0941/US\$

- PPP spot exchange rate (\$A/\$B) = Initial spot rate $\times \left[\frac{1 + \text{Inflation}_A}{1 + \text{Inflation}_B} \right]^t$

Where: t = periods into the future

$$\Rightarrow \text{PPP spot exchange rate} = 1.0941 \times \frac{1 + 3.13\%}{1 + 1.15\%} = \text{A\$1.11552/US\$}$$

- We have:

+ Price of 3-piece luggage set in US\$ = \$850

+ PPP spot exchange rate = A\$1.11552/US\$

$$\Rightarrow \text{Price of 3-piece luggage set in Sydney (A\$)} = \text{US\$850} \times \text{A\$1.11552/US\$} = \text{A\$948.192}$$

=> However, purchasing power parity is not always an accurate predictor of exchange rate movements, particularly in the short-term.

Problem 21: The Beer Standard. (DONE)

SOLVE

Note:

1. Beer price in South African rand = Price in local currency / Spot rate vs rand on 3/15/99
2. Implied PPP exchange rate = Price in local currency / Price in South Africa (rand)
= Price in local currency / 2.30
3. Under or overvalued to rand = Implied PPP rate / Spot rate on 3/15/99

Beer Prices						
Country	Beer	Local currency	Price in currency	In rand	Implied PPP rate	Spot rate vs Rand (3/15/99)
South	Castle	Rand	2.30		-	-

Africa						
Botswana	Castle	Pula	2.20	= 2.20/0.75 = Rand 2.933	= 2.20/2.30 = Pula 0.96/Rand	0.75 = 1/0.75 = Rand 1.33/Pula
Ghana	Star	Cedi	1,200	= 1,200/379.10 = Rand 3.1654	= 1,200/2.30 = Cedi 521.74/Rand	379.10 = 1/379.10 = Rand 0.0024/Cedi
Kenya	Tusker	Shilling	41.25	= 41.25/10.27 = Rand 4.01655	= 41.25/2.30 = Shilling 17.935/Rand	10.27 = 1/10.27 = Rand 0.0974/Shilling
Malawi	Carlsberg	Kwacha	18.50	= 18.50/6.96 = Rand 2.6580	= 18.50/2.30 = Kwacha 8.0435/Rand	6.96 = 1/6.96 = Rand 0.144/Kwacha
Mauritius	Phoenix	Rupee	15.00	= 15.00/4.03 = Rand 3.722	= 15/2.30 = Rupee 6.522/Rand	4.03 = 1/4.03 = Rand 0.2481/Rupee
Namibia	Windhoek	N\$	2.50	= 2.50/1.00 = Rand 2.50	= 2.50/2.30 = N\$1.087/Rand	1.00 = 1/1 = Rand 1/N\$
Zambia	Castle	Kwacha	1,200	= 1,200/340.68 = Rand 3.5224	= 1,200/2.30 = Kwacha 521.74/Rand	340.68 = 1/340.68 = Rand 0.00293/Kwacha
Zimbabwe	Castle	Z\$	9.00	= 9/6.15 = Rand 1.4634	= 9/2.30 = Z\$ 3.913/Rand	6.15 = 1/6.15 = Rand 0.1626/N\$

Country	Beer	Implied PPP rate	Spot rate vs Rand (3/15/99)	Under or overvalued to Rand (%)
South Africa	Castle	-	-	
Botswana	Castle	= 2.20/2.30 = Pula 0.96/Rand	0.75	= (0.96 - 0.75)/0.75 = 28%
Ghana	Star	= 1,200/2.30 = Cedi 521.74/Rand	379.10	= (521.74 - 379.10)/379.10 = 37.626%
Kenya	Tusker	= 41.25/2.30 = Shilling 17.935/Rand	10.27	= (17.935 - 10.27)/10.27 = 74.635%

Malawi	Carlsberg	= 18.50/2.30 = Kwacha 8.0435/Rand	6.96	= (8.0435 - 6.96)/6.96 = 15.57%
Mauritius	Phoenix	= 15/2.30 = Rupee 6.522/Rand	4.03	= (6.522 - 4.03)/4.03 = 61.84%
Namibia	Windhoek	= 2.50/2.30 = N\$1.087/Rand	1.00	= (1.087 - 1)/1 = 8.7%
Zambia	Castle	= 1,200/2.30 = Kwacha 521.74/Rand	340.68	= (521.74 - 340.68)/340.68 = 53.15%
Zimbabwe	Castle	= 9/2.30 = Z\$ 3.913/Rand	6.15	= (3.913 - 6.15)/6.15 = -36.37%

Problem 23: AvtoVAZ of Russia's Kalina Export Pricing Analysis. (DONE)

SOLVE

- **Spot exchange rate = Rubles 30/US\$**

Calendar Year	2001	2002	2003	2004	2005	2006
a. Kalina Price with Russian inflation (rubles)	260,000	= 260,000(1 + 14%) = 296,400	= 296,400(1 + 12%) = 331,968	= 331,968(1 + 11%) = 368,484.48	= 368,484.48(1 + 8%) = 397,963.24	= 397,963.24(1 + 8%) = 429,800.3
b. Exchange rate (rubles = \$1.00) if PPP holds	Rubles 30/US\$	= 30 x (1 + 14%)/(1 + 2.5%) = Rubles 33.366/US\$	= 33.366 x (1 + 12%)/(1 + 3%) = Rubles 36.281/US\$	= 36.281 x (1 + 11%)/(1 + 3%) = Rubles 39.1/US\$	= 39.1 x (1 + 8%)/(1 + 3%) = Rubles 40.99/US\$	= 40.99 x (1 + 8%)/(1 + 3%) = Rubles 42.99/US\$
c. Export price if using PPP (dollars)	= Rubles 260,000 / (Rubles 30/US\$) = \$8,666.67	= Rubles 296,400 / (Rubles 33.366/US\$) = \$8,883.3	= Rubles 331,968 / (Rubles 36.281/US\$) = \$9,149.91	= Rubles 368,484.48 / (Rubles 39.1/US\$) = \$9,424.155	= Rubles 397,963.24 / (Rubles 40.99/US\$) = \$9,708.79	= Rubles 429,800.3 / (Rubles 42.99/US\$) = \$9,997.68
d. Export price at fixed exchange rate (dollars)	= \$8,666.67	= Rubles 296,400 / (Rubles 30/US\$) = US\$9,880	= Rubles 331,968 / (Rubles 30/US\$) = US\$11,065.6	= Rubles 368,484.48 / (Rubles 30/US\$) = US\$12,282.816	= Rubles 397,963.24 / (Rubles 30/US\$) = US\$13,265.44	= Rubles 429,800.3 / (Rubles 30/US\$) = US\$14,326.67

e. Vlad, one of the newly hired pricing strategists, believes that prices of automobiles in both domestic and export markets will both increase with the rate of inflation, and that the ruble/dollar exchange rate will remain fixed. What would this imply or forecast for the future export price of the Kalina? (coi lại câu này khi ôn thi)

=> Vlad is actually not saying anything different than what questions c) and d) addressed. If the export price is based on the initial dollar price of \$8,666.67 then rising with dollar inflation, it reaches \$9,998.27 in 2006 (same as part c)). Alternatively, if the pricing follows the price in the domestic market, in rubles, rising with Russian inflation, it reaches Rubles 429,800 in 2006, and when converted to US dollars at an assumed fixed rate of exchange of Rubles 30 - \$1.00, the same \$14,326.68 as in part d).

	2001	2002	2003	2004	2005	2006
If export price rises at dollar inflation	= Rubles 260,000 / (Rubles 30/US\$) = \$8,666.67	= \$8,666.67 x (1 + 2.5%) = \$8,883.34	= \$8,883.3 x (1 + 3%) = \$9,149.84	= \$9,149.84 x (1 + 3%) = \$9,424.33	= 9,424.33 x (1 + 3%) = \$9,707.06	= \$9,707.06 x (1 + 3%) = \$9,998.27
						= \$9,998.27 x Rubles 30/US\$ = Rubles 299,948.1

f. Exporters generally would prefer that their own currency weakens over time versus the currency of the customer - making their product offering increasingly affordable (cheaper), and hopefully increasing sales volume. Since AvtoVAZ's costs are all in Russian rubles, earning a hard currency like the dollar which would be slowly strengthening against the ruble might increase profit margins (depending on what happens to costs over time from other factors).

If most of the competition in the target dollar markets were dollar-based manufacturers, their costs and prices might be rising with dollar inflation. The answers to parts c) and d) provide some ideas or possible boundaries on what you might consider. At fixed exchange rates, the dollar price would rise quite high by 2006 (to \$14,326.68), whereas if the rate of exchange had remained fixed the export price would be much lower in 2006 (\$9,998.27). Of course, pricing strategies can and should be changed over time with changing market conditions, but the general consensus of analysis would be to expect to increase the at a rate somewhere in between c) and d) forecasts.

Ví dụ: Giải thích vì sao exporters thường prefer currency của họ devalue so với currency nước ngoài

For example, assume you are a U.S. exporter who sells widgets at \$10 each to a buyer in Europe. The exchange rate is €1=\$1.25. Therefore, the cost to your European buyer is €8 per widget. Now let's say the dollar weakens and the exchange rate is €1=\$1.35. Your buyer wants to negotiate a better price, and you can afford to give them a break while still clearing at least \$10 per widget. Even if you set the new price at €7.50 per widget, which is a 6.25% discount from your buyer's perspective, your price in dollars is \$10.13 at the current exchange rate. A weak U.S. dollar allows your export business to remain competitive in international markets. Conversely, a stronger currency can reduce export competitiveness and make imports cheaper, which can cause the trade deficit to widen further, eventually weakening the currency in a self-adjusting

mechanism. But before this happens, export-dependent industries can be damaged by an unduly strong currency.

CHAPTER 7: Foreign Currency Derivatives

Chapter 7: Problem 1, 2, 3, 4, and 9

- Ôn kĩ Problem 7.3 và 7.4

Problem 1: Saguaro Funds. (DONE)

Contract = 62,500 pounds

Maturity	Open	High	Low	Settle	Change	High	Open Interest
March	1.4246	1.4268	1.4214	1.4228	0.0032	1.4700	25,605
June	1.4164	1.4188	1.4146	1.4162	0.0030	1.4550	809

- If Tony buys 5 June pound futures, and the spot rate at maturity is \$1.3980/£, what is the value of her position?
- If Tony sells 12 March pound futures, and the spot rate at maturity is \$1.4560/£, what is the value of her position?
- If Tony buys 3 March pound futures, and the spot rate at maturity is \$1.4560/£, what is the value of her position?
- If Tony sells 12 June pound futures, and the spot rate at maturity is \$1.3980/£, what is the value of her position?

SOLVE

Assumptions	a)	b)	c)	d)
	June (BUY)	March (SELL)	March (BUY)	June (SELL)
Number of pounds per futures contract	62,500 pounds/contract	62,500 pounds/contract	62,500 pounds/contract	62,500 pounds/contract
Numbers of contract	5 pound futures	12 pound futures	3 pound futures	12 pound futures
Spot at maturity	\$1.3980/£	\$1.4560/£	\$1.4560/£	\$1.3980/£
Future	\$1.4162/£	\$1.4228/£	\$1.4228/£	\$1.4162/£
Gain (Loss)	$= (1.3980 - 1.4162) \times (62,500 \times 5)$	$= - (1.4560 - 1.4228) \times (62,500 \times 12)$	$= (1.4560 - 1.4228) \times (62,500 \times 3)$	$= (1.3980 - 1.4162) \times (62,500 \times 12)$

	= -5,687.5	= -24,900	= 6,225	= 13,650
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Problem 2: Amber McClain. (DONE)

EXHIBIT 7.1

Mexican Peso (CME) (MXN 500,000; \$ per 10MXN)

Maturity	Open	High	Low	Settle	Change	Lifetime		Open Interest
						High	Low	
Mar	0.10953	0.10988	0.10930	0.10958	...	0.11000	0.09770	34,481.00
June	0.10790	0.10795	0.10778	0.10773	...	0.10800	0.09730	3,405.00
Sept	0.10615	0.10615	0.10610	0.10573	...	0.10615	0.09930	1,481.00

All contracts are for 500,000 Mexican pesos. "Open" means the opening price on the day. "High" means the high price on the day. "Low" indicates the lowest price on the day. "Settle" is the closing price on the day. "Change" indicates the change in the settle price from the previous day's close. "High" and "Low" to the right of "Change" indicate the highest and lowest prices this specific contract (as defined by its maturity) has experienced over its trading history. "Open Interest" indicates the number of contracts outstanding.

- What is the value of her position at maturity if the ending spot rate is \$0.12000/Ps?
- What is the value of her position at maturity if the ending spot rate is \$0.09800/Ps?
- What is the value of her position at maturity if the ending spot rate is \$0.11000/Ps?

SOLVE

Assumptions	a)	b)	c)
	Values	Values	Values
Number of pesos per futures contract	500,000	500,000	500,000
Number of contract	8 futures contracts	8 futures contracts	8 futures contracts
Should she sell the peso futures?	Sell	Sell	Sell
Ending spot rate (\$/peso)	\$0.12000/Ps	\$0.09800/Ps	\$0.11000/Ps
Futures settle price from Exhibit 7.1 (\$/peso)	\$0.10773/Ps	\$0.10773/Ps	\$0.10773/Ps
Spot - Futures	= \$0.12000/Ps - \$0.10773/Ps = \$0.01227/Ps	= \$0.09800/Ps - \$0.10773/Ps = -\$0.00973/Ps	= \$0.11000/Ps - \$0.10773/Ps = \$0.00227/Ps
Value of total position at maturity (US\$)	= - (\$0.01227/Ps x 8) x 500,000 = -\$49,080/Ps	= (\$0.00973/Ps x 8) x 500,000 = \$38,920/Ps	= - (\$0.00227/Ps x 8) x 500,000 = -\$9,080/Ps

- **Interpretation:**

Amber buys at the spot price and sells at the futures price. If the futures price is greater than the ending spot price, she makes a profit.

Problem 3: Cece Cao in Jakarta. (HQC KI)

Option choices on the Singapore dollar:

Option choices on the Singapore dollar	Call on S\$	Put on S\$
Strike price (US\$/Singapore dollar)	\$0.6500	\$0.6500
Premium (US\$/Singapore dollar)	\$0.00046	\$0.00003

Assumptions	Values
Current spot rate (US\$/Singapore dollar)	\$0.6000
Days to maturity	90
Expected spot rate in 90 days (US\$/Singapore dollar)	\$0.700

SOLVE

a. Should Cece buy a put on Singapore dollars or a call on Singapore dollars?

- Since Cece expects the Singapore dollar to appreciate versus the US dollar, she should buy a call on Singapore dollars. This gives her the right to BUY Singapore dollars at a future date at \$0.65 each, and then immediately resell them in the open market at \$0.70 each for a profit. (If her expectation of the future spot rate proves correct.)

b. What is Cece's breakeven price on the option purchased in part a)?

- Note: This does not include any interest cost on the premium.

	Per S\$
Strike price	\$0.65000
Plus: Premium	\$0.00046
Break-even Price	$= \$0.65000 + \$0.00046 = \$0.65046$

c. What is Cece's gross profit and net profit (including premium) if the ending spot rate is \$0.70/S\$?

	Gross Profit (US\$/S\$)	Net Profit (US\$/S\$)
Spot rate	\$0.70000	\$0.70000
Less: Strike price	(\$0.65000)	(\$0.65000)
Less: Premium		(\$0.00046)
Profit	\$0.05000	\$0.04954

d. What is Cece's gross profit and net profit (including premium) if the ending spot rate is \$0.80/S\$?

	Gross Profit (US\$/S\$)	Net Profit (US\$/S\$)
Spot rate	\$0.80000	\$0.80000
Less: Strike price	(\$0.65000)	(\$0.65000)
Less: Premium		(\$0,00046)
Profit	\$0.15000	\$0.14954

Problem 4: Kapinsky Capital (A). (Cần ôn lại) (Học kĩ)

Assumptions	a)	b)
	Values	Values
Initial Investment (funds available)	\$10,000,000	\$10,000,000
Current spot rate (US\$/€)	\$1.3358	\$1.3358
30-day forward rate (US\$/€)	\$1.3350	\$1.3350
Expected spot rate in 30 days (US\$/€)	\$1.3600	\$1.2800

SOLVE

a. If Christoph believes the euro will continue to rise in value against the U.S. dollar, so that he expects the spot rate to be \$1.3600/€ at the end of 30 days, what should he do?

- One of the more interesting dimensions of speculating in the forward market, is that if the speculator has access to the forward market (bank lines or relationships when working on behalf of an established firm), many forward speculation strategies require no actual cash flow position up-front. In this case, Christoph believes the dollar will be trading at \$1.36/€ in the open market at the end of 30 days, but he has the ability to buy or sell dollars at a forward rate of \$1.3350/€. He should therefore buy euros forward 30 days (requires no actual cash flow up-front), and at the end of 30 days take delivery of those euros and sell in the spot market at the higher dollar rate for profit.

- Euros bought forward = $\frac{\text{Investment}}{\text{Forward rate}} = \frac{\$10,000,000}{\$1.3350/\text{€}} = \text{€}7,490,636.704$

- We have: Spot rate in open market at the end of 30 days = \$1.3600/€

=> US\$ proceeds (Euros bought forward exchanged to US\$ spot rate at the end of 30 days) is:

$$= \text{Euros bought forward} \times \text{Spot rate in 30 days} = \text{€}7,490,636.704 \times \$1.3600/\text{€} \\ = \$10,187,265.91744$$

=> Profits in US\$ = \$10,187,265.91744 - \$10,000,000 = \$187,265.91744

b. If Christoph believes the euro will depreciate in value against the U.S. dollar, so that he expects the spot rate to be \$1.2800/€ at the end of 30 days, what should he do?

Again, a profitable strategy can be executed without any actual cash flow changing hands at the beginning of the period. Since Christoph believes that the dollar will strengthen to \$1.28 in 30 days, he should sell euros forward now at the higher dollar rate, wait 30 days and buy the euros needed on the open market at \$1.28, and immediately then use those euros to fulfill his forward contract to sell euros for dollars at \$1.3350 for a profit.

- Euros bought in open market in 30 days = $\frac{Investment}{Spot\ rate} = \frac{\$10,000,000}{\$1.28/\text{€}} = \text{€}7,812,500$

Because Stefan had sold these euros forward at the start of the 30 day period thanks

- US\$ proceeds (Euros sold forward into US\$) is:
 = Euros bought in open market in 30 days x 30-day forward rate (US\$/€)
 = €7,812,500 x \$1.3350/€ = \$10,429,687.50

=> Profits in US\$ = \$10,429,687.50 - \$10,000,000 = \$429,687.50

Problem 9: Calling All Profits. (Cần ôn lại)

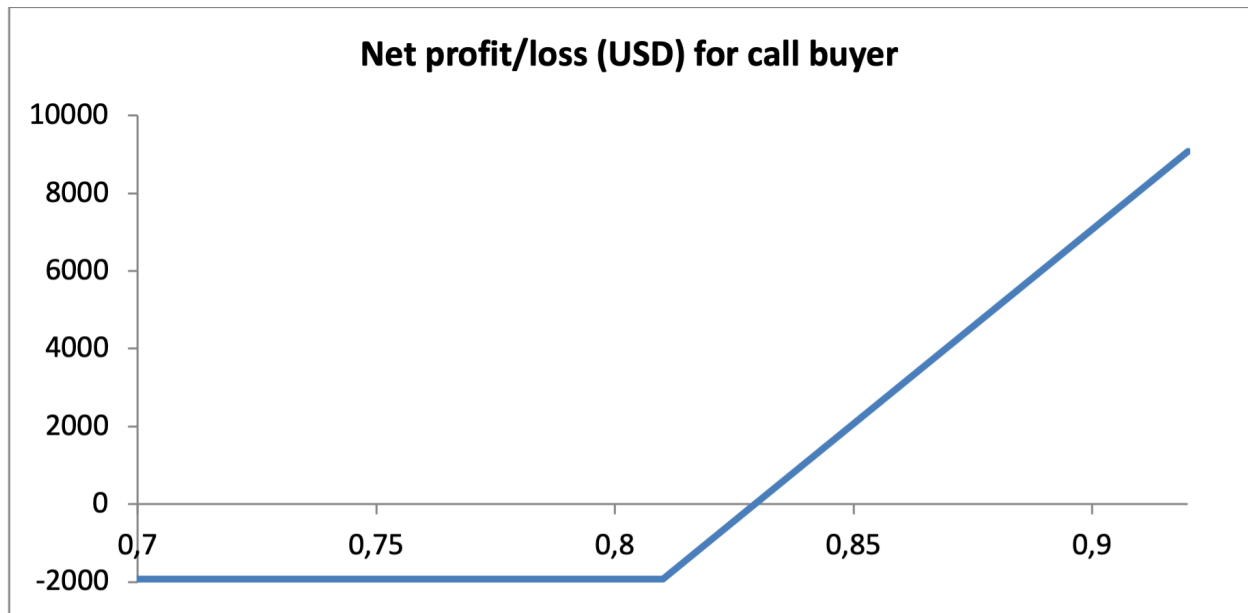
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Assumptions	Values
Notional principal (NZD)	NZD 100,000
Strike price (USD/NZD)	USD 0.8100
Premium (USD/NZD)	USD 0.01920

SOLVE

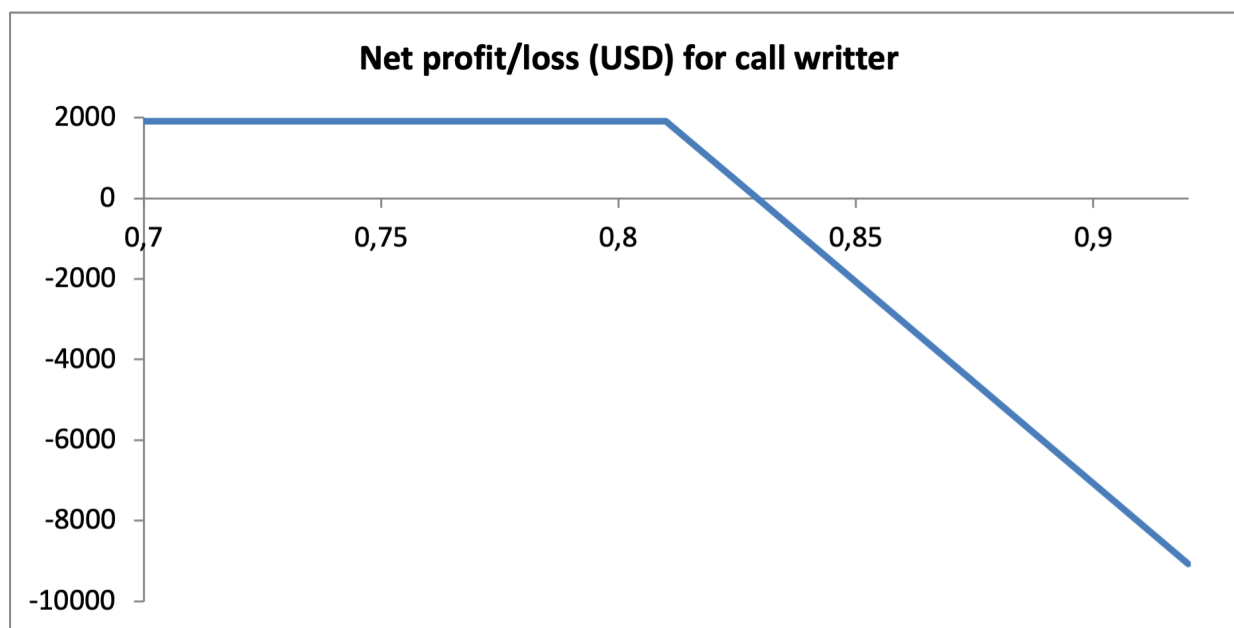
a. Suppose that you have bought such a call option. Plot your profit or loss on a graph should you exercise before maturity at a time when the NZ\$ is traded spot at between \$0.7000/NZ\$ and \$0.9200/NZ\$. Find the break-even exchange rate.

		Strike price	Break-even	
Spot rate (USD/NZD)	0.7000	0.8100	= 0.8100 + 0.0192 = 0.8292	0.92
Call payoff (USD)	= NZD100,000 x Max(0; 0.70 - 0.81) = \$0	= NZD100,000 x Max(0; 0.81 - 0.81) = \$0	= NZD100,000 x Max(0; 0.8292 - 0.8100) = \$1,920	= NZD100,000 x Max (0; 0.92 - 0.81) = \$11,000
Less: Option premium (USD)	= NZD100,000 x \$0.01920/NZD = \$1,920	= NZD100,000 x \$0.01920/NZD = \$1,920	= NZD100,000 x \$0.01920/NZD = \$1,920	= NZD100,000 x \$0.01920/NZD = \$1,920
Net profit/loss (USD)	= \$0 - \$1,920 = -\$1,920	= \$0 - \$1,920 = -\$1,920	= \$1,920 - \$1,920 = \$0	= \$11,000 - \$1,920 = \$9,080



b. Repeat (a) if you have sold such a call option

		Strike price	Break-even	
Spot rate (USD/NZD)	0.7000	0.8100	$= 0.8100 + 0.0192$ $= 0.8292$	0.92
Option premium (USD)	$= \text{NZD } 100,000 \times \$0.0192/\text{NZD}$ $= \$1,920$	$= \text{NZD } 100,000 \times \$0.0192/\text{NZD}$ $= \$1,920$	$= \text{NZD } 100,000 \times \$0.0192/\text{NZD}$ $= \$1,920$	$= \text{NZD } 100,000 \times \$0.0192/\text{NZD}$ $= \$1,920$
Less: Call payoff (USD)	$= \text{NZD } 100,000 \times \text{Max}(0; 0.70 - 0.81)$ $= \$0$	$= \text{NZD } 100,000 \times \text{Max}(0; 0.81 - 0.81)$ $= \$0$	$= \text{NZD } 100,000 \times \text{Max}(0; 0.8292 - 0.81)$ $= \$1,920$	$= \text{NZD } 100,000 \times \text{Max}(0; 0.92 - 0.81)$ $= \$11,000$
Net profit/loss (USD)	$= \$1,920 - \0 $= \$1,920$	$= \$1,920 - \0 $= \$1,920$	$= \$1,920 - \$1,920$ $= \$0$	$= \$1,920 - \$11,000$ $= \$9,080$



CHAPTER 9: Foreign Exchange Rate Determination

Chapter 9: Problem 1, 2, 3, 4, and 5

Problem 1:

Problem 2:

CHAPTER 10: Transactions Exposure

Chapter 10: Problem 1, 2, 3, 4, 8 and 10

Problem 1: BioTron Medical, Inc. (DONE, đã hiểu)

Assumptions:

- BioTron's 30-day account receivable, Japanese yen = ¥10,000,000
- Spot rate, ¥/\$ = 107.91
- 30-day forward rate, ¥/\$ = 107.66
- 90-day forward rate, ¥/\$ = 106.81
- 180-day forward rate, ¥/\$ = 105.89
- Numata's WACC = 9.38%
- BioTron Medical's WACC = 7.85%
- Desired discount on purchase price by Numata = 5%

SOLVE

- Brent Bush should compare two basic alternatives, both of which eliminate the currency risk:

1. Allow the discount and receive payment in Japanese yen in cash:

Account receivable (yen)	¥10,000,000
Discount for cash payment up-front (5%)	-¥10,000,000 x 5% = (¥500,000)
Amount paid in cash net of discount	= ¥10,000,000 - ¥500,000 = ¥9,500,000

$$\Rightarrow \text{Amount received by BioTron (USD)} = \frac{\text{¥9,500,000}}{\text{¥107.91/\$}} = \$88,036.327$$

2. Not offer any discounts for early payment and cover exposure with forwards:

Account receivable (yen)	¥10,000,000
90-day forward rate	¥106.81/\$
Amount received in cash in dollars, in 90 days	$= \frac{\text{¥10,000,000}}{\text{¥106.81/\$}} = \$93,624.2$
Discount factor for 90 days with BioTron's WACC (tại sao chia 360 days)	$= \frac{1}{1 + (7.85\% \times 90/360)} = 0.980753$
Present value of dollar cash received	$= \$93,624.2 \times 0.980753 = \$91,821.99$

$\Rightarrow$ Brent Bush should politely decline Numata's offer to pay cash in exchange for the requested discount (reject option 1 because $\$88,036.327 < \$91,821.99$).

Problem 2: Bobcat Company. (Học kĩ)

Assumptions:

- Purchase price of Korean manufacturer (KRW) = KRW7,800,000,000
- Spot rate, ¥/\$ = 107.91
- 30-day forward rate, ¥/\$ = 107.66
- 90-day forward rate, ¥/\$ = 106.81
- 180-day forward rate, ¥/\$ = 105.89
- Numata's WACC = 9.38%
- BioTron Medical's WACC = 7.85%
- Desired discount on purchase price by Numata = 5%

Assumptions	Values
Purchase price of Korean manufacturer (KRW)	7,800,000,000
Less: Initial payment (KRW)	(1,500,000,000)
Net settlement needed (KRW), in 6 months	6,300,000,000

Current spot rate (KRW/USD)	Won1,071.95/\$
Six month forward rate (KRW/USD)	Won1,103.28/\$
Bobcat's cost of capital (WACC)	9%

	United States	Korea
Six-month interest rate (investment rate)	5% per annum	12% per annum
Borrowing premium	1% per annum	1% per annum
Six-month borrowing rate	= 5% + 1% = 6% per annum	= 12% + 1% = 13% per annum

Options on KRW	Call Option	Put Option
Strike Price (KRW/USD)	Won1,100/\$	Won1,100/\$
Option Premium (%)	2.83%	2.48%

SOLVE

1. Remain Unhedged: Make won payment in 6 months at spot rate.

	Value of Payable (USD)	Risk Assessment
If spot rate in 6 months is same as current spot rate (Won1,071.95/\$)	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,071.95/\$}$ = \$5,877,139.8	
If spot rate in 6 months is same as forward rate (Won1,103.28/\$)	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,103.28/\$}$ = \$5,710,245.8125	
Possible spot rate in 6 months (KRW/USD)	Won1,100/\$	
Cost of settlement in 6 months	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,100/\$}$ = \$5,727,272.73	

2. Forward Market Hedge: Buy won forward 6 months

	Value of Payable (USD)	Risk Assessment
Forward in 6 months (KRW/USD)	Won1,100/\$	
Cost of settlement in 6 months	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,103.28/\$}$	

	= \$5,710,245.8125	
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3. Money Market Hedge: Exchange dollars for won now, invest for 6 months

	Value of Payable (USD)	Risk Assessment
Discount factor at 6 month KRW interest rate	$= \frac{1}{(1 + 12\% \times 180/360)}$ = 0.9434	
Present value of payable in KRW	= Won6,300,000,000 x 0.9434 = Won5,943,420,000	
Present value of payable in USD	$= \frac{\text{Won}5,943,420,000}{\text{Won}1,071.95/\$}$ = \$5,544,493.68	
WACC carry-forward factor for 6 months (WHYYYY???)	$= 1 + 9\% \times \frac{180}{360} = 1.045$	
Cost of settlement in 6 months	= \$5,544,493.68 x 1.045 = \$5,793,995.896	

4. Option Market Hedge: Buy call on won

	Value of Payable (USD)	Risk Assessment
Option principal (KRW)	Won6,300,000,000	
Option premium in USD	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,071.95/\$} \times 2.83\%$ = \$166,323.056	
WACC carry-forward factor for 6 months	$= 1 + 9\% \times \frac{180}{360} = 1.045$	
Value of option premium in 6 months	= \$166,323.056 x 1.045 = \$173,087.594	
Spot rate (KRW/USD)		
Exercise call option		
Cost of settlement in 6 months		

Problem 3: Siam Cement. (Học kĩ)

Assumptions:

- US dollar debt taken out in June 1997 = \$50,000,000
- US dollar borrowing rate on debt = 8.4%
- Initial spot exchange rate, baht/dollar, June 1997 = 25.0
- Average spot exchange rate, baht/dollar, June 1998 = 42.0

SOLVE

Calculation of Foreign Exchange Loss on Repayment of Loan:

At the time the loan was acquired, the scheduled repayment of dollar and baht amounts would have been as follows:

- **Scheduled Repayment:**

- Repayment of US dollar debt: Principal = \$50,000,000
- Repayment of US dollar debt: Interest = \$50,000,000 x 8.4% = \$4,200,000

=> Total repayment = \$50,000,000 + \$4,200,000 = \$54,200,000

- Exchange rate at time of repayment, baht/dollar = B25.0/\$

=> Total repayment in Thai baht = \$54,200,000 x B25.0/\$ = Baht 1,355,000,000

- Total proceeds from loan, up-front, in Thai baht = \$50,000,000 x B25.0/\$ = Baht 1,250,000,000

=> Net interest to be paid, in Thai baht = Baht 1,355,000,000 - Baht 1,250,000,000 = Baht 105,000,000

- **Actual Repayment:**

- Repayment of US dollar debt: Principal = \$50,000,000
- Repayment of US dollar debt: Interest = \$50,000,000 x 8.4% = \$4,200,000

=> Total repayment = \$50,000,000 + \$4,200,000 = \$54,200,000

- Exchange rate at time of repayment, baht/dollar = B42.0/\$

=> Total repayment in Thai baht = \$54,200,000 x B42.0/\$ = Baht 2,276,400,000

- Total proceeds what Siam had EXPECTED or SCHEDULED to be repaid = \$54,200,000 x B25.0/\$
= Baht 1,355,000,000

=> Amount of foreign exchange loss on debt = Baht 2,276,400,000 - Baht 1,355,000,000
= Baht 921,400,000

Problem 4: P&G India.

Assumptions:

- 180-day account payable (VND) = 10,000,000
- Spot rate, VND/INR = 346.49
- 180-day forward rate, VND/INR = 338.28
- Expected spot rate in 180 days (VND/INR) = 339.86
- 180-day INR investing rate = 6.0%
- 180-day VND investing rate = 1.80%
- Currency agent's exchange rate fee = 3.85%
- P&G India's cost of capital = 10.0%

SOLVE

1. Remain Unhedged: Settle payable in 180 days at spot rate.

	Value of Payable (USD)	Risk Assessment
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	$\begin{aligned} &= \frac{\text{Won}6,300,000,000}{\text{Won}1,071.95/\$} \\ &= \$5,877,139.8 \end{aligned}$	
	$\begin{aligned} &= \frac{\text{Won}6,300,000,000}{\text{Won}1,103.28/\$} \\ &= \$5,710,245.8125 \end{aligned}$	
	Won1,100/\$	
	$\begin{aligned} &= \frac{\text{Won}6,300,000,000}{\text{Won}1,100/\$} \\ &= \$5,727,272.73 \end{aligned}$	

2. Forward Market Hedge: Buy 180 days forward on VND

	Value of Payable (USD)	Risk Assessment
	Won1,100/\$	
	$\begin{aligned} &= \frac{\text{Won}6,300,000,000}{\text{Won}1,103.28/\$} \\ &= \$5,710,245.8125 \end{aligned}$	

3. Money Market Hedge:

	Value of Payable (USD)	Risk Assessment
	$\begin{aligned} &= \frac{1}{(1 + 12\%/2)} \\ &= 0.9434 \end{aligned}$	
	$\begin{aligned} &= \text{Won}6,300,000,000 \times 0.9434 \\ &= \text{Won}5,943,420,000 \end{aligned}$	
	$\begin{aligned} &= \frac{\text{Won}5,943,420,000}{\text{Won}1,071.95/\$} \\ &= \$5,544,493.68 \end{aligned}$	
	$= 1 + \frac{9\%}{2} = 1.045$	
	$\begin{aligned} &= \$5,544,493.68 \times 1.045 \\ &= \$5,793,995.896 \end{aligned}$	

4. Indian Currency Agent Hedge

	Value of Payable (USD)	Risk Assessment
Option principal (KRW)	Won6,300,000,000	
Option premium in USD	$= \frac{\text{Won}6,300,000,000}{\text{Won}1,071.95/\$} \times 2.83\%$ = \$166,323.056	
WACC carry-forward factor for 6 months	$= 1 + \frac{9\%}{2} = 1.045$	
Value of option premium in 6 months	$= \$166,323.056 \times 1.045$ = \$173,087.594	
Spot rate (KRW/USD)		
Exercise call option		
Cost of settlement in 6 months		

CHAPTER 13: Global Cost of Capital

Problem 1: Ganado's Cost of Capital. (DONE)

Assumptions	Domestic CAPM	International ICAPM
Ganado's beta, β	1.05	0.85
Risk-free rate of interest, k_{rf}	3.60%	3.60%
Company credit risk premium	4.40%	4.40%
Cost of debt, before tax, k_d	8.00%	8.00%
Corporate income tax rate, t	35%	35%
General return on market portfolio, k_m	9.00%	8.00%
Optimal capital structure: Proportion of debt, D/V Proportion of equity, E/V	30% 70%	30% 70%

SOLVE

a) Ganado's cost of equity:

Cost of Equity	Domestic CAPM	International CAPM
$k_e = k_{rf} + (k_m - k_{rf})\beta$	$k_e = 3.60\% + (9.00\% - 3.60\%) \times 1.05$ $= 9.27\%$	$k_e = 3.60\% + (8.00\% - 3.60\%) \times 0.85$ $= 7.34\%$

b) Ganado's cost of equity, after tax:

Cost of Debt	Domestic CAPM	International CAPM
$k_d (\text{After tax}) = k_d \times (1 - t)$	$k_d = 8\% \times (1 - 35\%) = 5.2\%$	$k_d = 8\% \times (1 - 35\%) = 5.2\%$

c) Ganado's weighted average cost of capital:

Cost of Equity	Domestic CAPM	International CAPM
$WACC = (k_e \times E/V) + [(k_d \times (1-t)) \times D/V]$	$WACC = (9.27\% \times 70\%) + (5.2\% \times 30\%)$ $= 8.049\%$	$WACC = (7.34\% \times 70\%) + (5.2\% \times 30\%)$ $= 6.698\%$

Problem 2: Ganado and Equity Risk Premiums. (DONE)

Assumptions	Domestic CAPM	International ICAPM
Ganado's beta, β	1.2	0.90
Risk-free rate of interest, k_{rf}	4.00%	4.00%
Company credit risk premium		
Cost of debt, before tax, k_d	8.00%	8.00%
Corporate income tax rate, t	35%	35%
Equity risk premium	8.00%	8.00%
General return on market portfolio, k_m	9.00%	8.00%
Optimal capital structure: Proportion of debt, D/V Proportion of equity, E/V	40% 60%	40% 60%

SOLVE

a) Ganado's cost of equity: $k_e = k_{rf} + (k_m - k_{rf})\beta$

Equity risk premium = 8.00%	Equity risk premium = 7.00%
-----------------------------	-----------------------------

Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
$= 4\% + 8\% \times 1.2$ $= 13.6\%$	$= 4\% + 8\% \times 0.90$ $= 11.2\%$	$= 4\% + 7\% \times 1.2$ $= 12.4\%$	$= 4\% + 7\% \times 0.90$ $= 10.3\%$

Equity risk premium = 5.00%		Equity risk premium = 4.00%	
Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
$= 4\% + 5\% \times 1.2$ $= 10\%$	$= 4\% + 5\% \times 0.90$ $= 8.5\%$	$= 4\% + 4\% \times 1.2$ $= 8.8\%$	$= 4\% + 4\% \times 0.90$ $= 7.6\%$

b) Ganado's cost of debt, after tax:

Equity risk premium = 8.00%		Equity risk premium = 7.00%	
Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$

Equity risk premium = 5.00%		Equity risk premium = 4.00%	
Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$	$= 8\% \times (1 - 35\%)$ $= 5.2\%$

c) Ganado's weighted average cost of capital: $WACC = (k_e \times E/V) + [(k_d \times (1-t)) \times D/V]$

Equity risk premium = 8.00%		Equity risk premium = 7.00%	
Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
$= 13.6\% \times 60\% +$ $5.2\% \times 40\%$ $= 10.24\%$	$= 11.2\% \times 60\% +$ $5.2\% \times 40\%$ $= 8.8\%$	$= 12.4\% \times 60\% +$ $5.2\% \times 40\%$ $= 9.52\%$	$= 10.3\% \times 60\% +$ $5.2\% \times 40\%$ $= 8.26\%$

Equity risk premium = 5.00%		Equity risk premium = 4.00%	
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Domestic CAPM	International CAPM	Domestic CAPM	International CAPM
= 10% x 60% + 5.2% x 40% = 8.08%	= 8.5% x 60% + 5.2% x 40% = 7.18%	= 8.8% x 60% + 5.2% x 40% = 7.36%	= 7.6% x 60% + 5.2% x 40% = 6.64%

Problem 3: Pharos Papers. (DONE)

Assumptions	a) Values	b) Values
Pharos' beta	1.20	0.75
Cost of debt, before tax	12%	12%
Risk-free rate of interest	5%	5%
Corporate income tax rate	20%	20%
General return on market portfolio	9%	9%
Optimal capital structure:		
Proportion of debt, D/V	65%	65%
Proportion of equity, E/V	35%	35%

SOLVE

- **Ganado's cost of equity:** $k_e = k_{rf} + (k_m - k_{rf})\beta$

	a)	b)
Cost of debt, after tax $k_d (After\ tax) = k_d \times (1 - t)$	= 12% x (1 - 20%) = 9.6%	= 12% x (1 - 20%) = 9.6%
Cost of equity, after tax $k_e = k_{rf} + (k_m - k_{rf})\beta$	= 5% + 1.20 x (9% - 5%) = 9.8%	= 5% + 0.75 x (9% - 5%) = 8%
WACC = $(k_e \times E/V) + [(k_d \times (1-t)) \times D/V]$	= 9.8% x 35% + 9.6% x 65% = 9.67%	= 8% x 35% + 9.6% x 65% = 9.04%

Problem 4: Nestle of Switzerland Revisited. (DONE) (Học kĩ)

Assumptions	a) Values	b) Values
Swiss bond index yield, the risk-free rate	0.52%	0.52%
Swiss equity market return, in Swiss francs	8.4%	
Global equity yield, in Swiss francs		8.820%
Nestle's beta versus Swiss equity market	0.825	
Nestle's beta versus Global equity market		0.515

SOLVE

	Domestic Portfolio	Global Portfolio
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Nestle's cost of equity using CAPM $k_e = k_{rf} + \beta(k_m - k_{rf})$	$= 0.52\% + 0.825 \times (8.4\% - 0.52\%)$ $= 7.021\%$	$= 0.52\% + 0.515 \times (8.820\% - 0.52\%)$ $= 4.7945\%$
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Problem 5: EuroAsia AutoPart Firm. (DONE)

Assumptions	a) Values	b) Values
Eurasia's beta	1.1	0.9
Cost of debt, before tax	16%	16%
Risk-free rate of interest	8.5%	8.5%
Corporate income tax rate	25%	25%
General return on market portfolio	22%	22%
Optimal capital structure:		
Proportion of debt, D/V	80%	80%
Proportion of equity, E/V	20%	20%

SOLVE

- **Calculate WACC:**

	a)	b)
Cost of debt, after tax $k_d (\text{After tax}) = k_d \times (1 - t)$	$= 16\% \times (1 - 25\%)$ $= 12\%$	$= 16\% \times (1 - 25\%)$ $= 12\%$
Cost of equity, after tax $k_e = k_{rf} + (k_m - k_{rf})\beta$	$= 8.5\% + 1.1 \times (22\% - 5\%)$ $= 27.2\%$	$= 8.5\% + 0.9 \times (22\% - 5\%)$ $= 23.8\%$
WACC $= (k_e \times E/V) + [(k_d \times (1-t)) \times D/V]$	$= 12\% \times 80\% + 27.2\% \times 20\%$ $= 15.04\%$	$= 12\% \times 80\% + 23.8\% \times 20\%$ $= 14.36\%$

Problem 11: Genedak-Hogan's WACC. (DONE)

Assumptions	Symbol	Before	After
		Diversification	Diversification
Correlation between G-H and the market	ρ_{jm}	0,88	0,76
Standard deviation of G-H's returns	σ_j	28,0%	26,0%
Standard deviation of market's returns	σ_m	18,0%	18,0%
Risk-free rate of interest	k_{rf}	3,0%	3,0%
Additional equity risk premium for internationalization	RPM	0,0%	3,0%

Estimate of G-H's cost of debt in US market	kd	7,2%	7,0%
Market risk premium	km-krf	5,5%	5,5%
Corporate tax rate	t	35,0%	35,0%
Proportion of debt	D/V	38%	32%
Proportion of equity	E/V	62%	68%

SOLVE

● Estimating Cost of Capital:

	Before Diversification	After Diversification
Estimated beta $\beta = (\rho_{jm} \times \sigma_j) / (\sigma_m)$	$= (0.88 \times 28\%) / 18\%$ $= 1.369$	$= (0.76 \times 26\%) / 18\%$ $= 1.098$
Estimated cost of equity $k_e = k_{rf} + \beta(k_m - k_{rf})$	$= 3\% + 1.369 \times 5.5\%$ $= 10.529\%$	$= 3\% + 1.098 \times 5.5\%$ $= 9.04\%$
Estimated cost of equity with additional risk premium $k_e^* = k_{rf} + \beta(k_m - k_{rf}) + \text{RPM}$	$= 10.529\% + 3\%$ $= 10.529\%$	$= 9.04\% + 3\%$ $= 12.04\%$
Cost of debt, after tax $k_d (\text{After tax}) = k_d \times (1 - t)$	$= 7.2\% \times (1 - 35\%)$ $= 4.68\%$	$= 7\% \times (1 - 35\%)$ $= 4.55\%$
Weighted average cost of capital $\text{WACC} = (k_e \times E/V) + [(k_d \times (1-t)) \times D/V]$	$= 10.529\% \times 62\% + 4.68\% \times 38\%$ $= 8.30638\%$	$= 9.04\% \times 68\% + 4.55\% \times 32\%$ $= 7.6032\%$
Weighted average cost of capital with additional risk premium $\text{WACC} = (k_e^* \times E/V) + [(k_d \times (1-t)) \times D/V]$	$= 10.529\% \times 62\% + 4.68\% \times 38\%$ $= 8.30638\%$	$= 12.04\% \times 68\% + 4.55\% \times 32\%$ $= 9.6432\%$

- There are a number of different factors at work here. First, as a result of international diversification, their access to debt has improved, resulting in a lower cost of debt capital. This is not fully appreciated, however, as the firm has chosen to reduce its overall use of debt post-diversification (common among MNEs).
- The firm's WACC does indeed drop for the standardized case. If, however, the market assessed an additional equity risk premium of 3.0%, the benefits are swamped by the higher required return on equity by the market.

CHAPTER 14: Raising Equity and Debt Globally

Problem 1: Al-Niger Co. (Nigeria). (DONE)

Assumptions	a) Depreciating €	b) Appreciating €
Principal borrowed (euros)	€10,000,000	€10,000,000
Euro interest rate, one year (% per annum)	8.0%	8.0%
Beginning of year spot rate, NGN/€	NGN225/€	NGN225/€
End of year spot rate, NGN/€	NGN200/€	NGN250/€

SOLVE

Calculation of Principal and Interest		
Euro-denominated debt, in Euros		
Principal	€10,000,000	€10,000,000
Interest	= €10,000,000 x 8% = €800,000	= €10,000,000 x 8% = €800,000
Principal and interest due at end of year	= €10,000,000 + €800,000 = €10,800,000	= €10,000,000 + €800,000 = €10,800,000

Cost of funds if euro depreciates versus NGN		
	a) Depreciating €	b) Appreciating €
Repayment cost of euros, in NGN (ending spot rate)	= €10,800,000 x NGN200/€ = NGN2,160,000,000	= €10,800,000 x NGN250/€ = NGN2,700,000,000
The NGN value of initial euro debt proceeds (at beginning of period spot rate)	= €10,000,000 x NGN225/€ = NGN2,250,000,000	= €10,000,000 x NGN225/€ = NGN2,250,000,000
NGN cost of euro (€) debt	$= \frac{NGN2,160,000,000}{NGN2,250,000,000} - 1$ = -0.04 = -4.0%	$= \frac{NGN2,700,000,000}{NGN2,250,000,000} - 1$ = 0.2 = 20%

Problem 2: Foreign Exchange Risk and the Cost of Swiss francs. (DONE)

Assumptions	Values
Debt principal (Swiss francs)	SF1,500,000
Initial spot rate	1.50

Cost of debt	5.0%
Tax rate	34.0%

SOLVE

a) If the exchange remains the same:

- The percentage change in SF the exchange rate, s:

$$s = \frac{S_1 - S_2}{S_2} \times 100 = \frac{1.50 - 1.50}{1.50} \times 100 = 0.00\%$$

- The effective cost of debt after exchange rate changes:

$$k_d = [(1 + k_d^{SF}) \times (1 + s)] - 1 = [(1 + 5.0\%) \times (1 + 0.00\%)] - 1 = 5.0\%$$

b) If the exchange ends the period at SF1.44/\$:

- The percentage change in SF the exchange rate, s:

$$s = \frac{S_1 - S_2}{S_2} \times 100 = \frac{1.50 - 1.44}{1.44} \times 100 = 4.167\%$$

- The effective cost of debt after exchange rate changes:

$$k_d = [(1 + k_d^{SF}) \times (1 + s)] - 1 = [(1 + 5.0\%) \times (1 + 4.167\%)] - 1 = 9.375\%$$

c) If the exchange ends the period at SF1.3860/\$:

- The percentage change in SF the exchange rate, s:

$$s = \frac{S_1 - S_2}{S_2} \times 100 = \frac{1.50 - 1.3860}{1.3860} \times 100 = 8.225\%$$

- The effective cost of debt after exchange rate changes:

$$k_d = [(1 + k_d^{SF}) \times (1 + s)] - 1 = [(1 + 5.0\%) \times (1 + 8.225\%)] - 1 = 13.636\%$$

d) If the exchange ends the period at SF1.6240/\$:

- The percentage change in SF the exchange rate, s:

$$s = \frac{S_1 - S_2}{S_2} \times 100 = \frac{1.50 - 1.6240}{1.6240} \times 100 = -7.6355\%$$

- The effective cost of debt after exchange rate changes:

$$k_d = [(1 + k_d^{SF}) \times (1 + s)] - 1 = [(1 + 5.0\%) \times (1 - 7.6355\%)] - 1 = -3.017\%$$

Problem 3: MAN Invest (UK). (DONE)

Assumptions	Values
Principal borrowed for 3 years, in euros	€100,000,000
Interest rate on loan, percent per annum	7.0%
Beginning spot rate, £/€	0.71
Expected % change in the euro versus the dollar	-8.0%

SOLVE

a) What is the effective cost of this loan for MAN Invest?

Calculation of the pound cost of euro debt	Year 0	Year 1	Year 2	Year 3
Proceeds from borrowing euros	€100,000,000			

Interest payment due, in euros		= €100,000,000 x 7% = -€7,000,000	= €100,000,000 x 7% = -€7,000,000	= €100,000,000 x 7% = -€7,000,000
Repayment of principal in year 3				-€100,000,000
Total cash flows of euro-denominated debt	€100,000,000	-€7,000,000	-€7,000,000	-€107,000,000

- **Note:** One way to check this calculation is to find the internal rate of return, IRR of this cash flow stream: This shows it is truly a 7.0% loan.

$$NPV = 0 = €100,000,000 - \frac{€7,000,000}{(1 + IRR)^1} - \frac{€7,000,000}{(1 + IRR)^2} - \frac{€107,000,000}{(1 + IRR)^3} \Leftrightarrow IRR = 7.0\%$$

	Year 0	Year 1	Year 2	Year 3
Expected exchange rate, £/€ = Spot rate x (1 - 8%)	£0.71/€	= £0.71/€ x (1 - 8%) = £0.6532/€	= £0.6532/€ x (1 - 8%) = £0.60/€	= £0.60/€ x (1 - 8%) = £0.553/€
British pound equivalent of euro-denominated cash flows	= £0.71/€ x €100,000,000 = £71,000,000	= £0.6532/€ x - €7,000,000 = -€4,572,400	= £0.60/€ x - €7,000,000 = -€4,200,000	= £0.553/€ x - €107,000,000 = €59,171,000

- **IRR of cash flow stream (the implied US dollar cost):**

$$NPV = 0 = £71,000,000 - \frac{€4,572,400}{(1 + IRR)^1} - \frac{€4,200,000}{(1 + IRR)^2} - \frac{€59,171,000}{(1 + IRR)^3}$$

$$\Leftrightarrow IRR = -1.556\%$$

- **NOTE:** The internal rate of return, IRR, of any stream of cash flows will determine the rate of discount (interest) which results in a NPV of the cash flow stream of exactly zero.

Problem 4: Inter-KSA (Saudi Arabia) (DONE)

Assumptions	Value
Working capital debt needed for one year	€50,000,000
Borrowing euros in Paris:	
Interest rate, percent per annum	7.0%
Principal borrowed	€50,000,000
Borrowing riyals in Saudi Arabia:	
Interest rate, percent per annum	9.0%
Initial spot rate, SR/Euro	SF4.5/€
Principal borrowed	SAR 225,000,000

SOLVE

Question: At what ending exchange rate would Inter-KSA be indifferent between borrowing euros and borrowing Saudi riyals?

- Cost of repaying the Saudi riyal loan in SR = SAR225,000,000 x (1 + 9.0%) = SAR245,250,000
- Cost of repaying the Euro in € = €50,000,000 x (1 + 7.0%) = €53,500,000

=> Breakeven exchange rate (SR cost/€ cost) = SAR245,250,000 / €53,500,000 = SAR 4.5841/€

CHAPTER 16: International Trade Finance

Problem 1: Nikken Microsystems (A). (Done)

Assumptions	Values
Face amount of sale	€700,000
Maturity, days	90
Trade acceptance fee, per annum	1.0%
Discount rate on sale of acceptance, per annum	4.0%

SOLVE

All-in-Cost of Trade Acceptance	
Face amount of the receivable	€700,000
Less: Trade acceptance fee <i>= Amount financed x Acceptance fee x (Days/360)</i>	= -[€700,000 x 1.0% x (90/360)] = (€1,750)
Less: Discount on the sale of acceptance <i>= Amount financed x Discount rate x (Days/360)</i>	= -[€700,000 x 4.0% x (90/360)] = (€7,000)
Net proceeds	= €700,000 - €1,750 - €7,000 = €691,250

$$\begin{aligned}
 \Rightarrow \text{Annualized percentage all-in-cost (AIC)} &= \frac{\text{Acceptance Fee} + \text{Discount}}{\text{Amount received}} \times \frac{360}{180} \\
 &= \frac{€1,750 + €7,000}{€691,250} \times \frac{360}{90} = 5.063\%
 \end{aligned}$$

Problem 2: Nikken Microsystems (B) (Cần ôn lại)

Assumptions	Values
Face amount of sale	€700,000
Maturity, days	90
Spot exchange rate, £/€	0.72
Forward exchange rate, 3-month, £/€	0.75
Trade acceptance fee, per annum	1.0%
Discount rate on sale of acceptance, per annum	4.0%

SOLVE

a. What are the British pound proceeds received at once?

Face amount of the receivable	€700,000
Less: Trade acceptance trade = <i>Face amount x Acceptance fee x (Days/360)</i>	= -[€700,000 x 1.0% x (90/360)] = (€1,750)
Euro proceeds	= €700,000 - €1,750 = €698,250
Spot exchange rate, £/€	£0.72/€
British pound proceeds, now	= €698,250 x £0.72/€ = £502,740

b. What are the pound proceeds received in 3 months under alternative 2?

Face amount of the receivable	€700,000
Less: Trade acceptance trade = <i>Face amount x Acceptance fee x (Days/360)</i>	= -[€700,000 x 1.0% x (90/360)] = (€1,750)
Euro net proceeds	= €700,000 - €1,750 = €698,250
3-month forward exchange rate, £/€	£0.75/€
British pound net proceeds received in 3-months	= €698,250 x £0.75/€ = £523,687.50

c. Breakeven reinvestment rate ?

British pounds received now, part a)	= £502,740
British pounds received at end of 90 days, part b)	= £523,687.50
Breakeven reinvestment rate of £ now to equal £ in 3 months (per annum)	= $\frac{£523,687.50 - £502,740}{£502,740} \times (360/90)$ = $\left(\frac{£523,687.50}{£502,740} - 1\right) \times (360/90)$ = 16.667%

d. Which alternative should Nikken Microsystem choose?

- If Nikken Microsystems' opportunity cost of capital is 16.667%, it should be indifferent financially between the two alternatives. However, selling the acceptance at once, alternative 1, improves Nikken's liquidity and removes the debt that otherwise would be financing the acceptance from Nikken Microsystem's balance sheet.

Problem 5: Alliasha - Toshiba. (Done)

Assumptions	Values
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Face amount of sale	€175,000
Down payment, 20% of payment	€35,000
Period for financing, days	90
Trade acceptance fee	3.5%
Discount rate on sale of acceptance, per annum	5.0%

SOLVE

a. What is the annualized percentage all-in-cost to Allisha-Toshiba?

Face amount of the receivable	€175,000
Less: Cash down payment	= - (€175,000 x 20%) = (€35,000)
Amount for financing	= €175,000 - €35,000 = €140,000
Less: Trade acceptance fee = <i>Amount financed x Acceptance fee x (Days/360)</i>	= - [€140,000 x 3.5% x (90/360)] = (€1,225)
Less: Discount for the period = <i>Amount financed x Acceptance fee x (Days/360)</i>	= - [€140,000 x 5% x (90/360)] = (€1,750)
Proceeds to Allisha-Toshiba	= €140,000 - €1,225 - €1,750 = €137,025

$$\Rightarrow \text{Annualized percentage all-in-cost (AIC)} = \frac{\text{Acceptance Fee} + \text{Discount}}{\text{Amount received}} \times \frac{360}{90}$$

$$= \frac{€1,225 + €1,750}{€137,025} \times \frac{360}{90} = 8.6845\%$$

b. What are Allisha-Toshiba's net cash proceeds, including the cash down payment?

- Total cash proceeds = Down payment + Proceeds of acceptance
= €35,000 + €137,025 = €172,025

Problem 7: BollyIndia Enterprises (A). (Cần ôn lại, được 90%) (HỌC KÌ)

Assumptions	Values
Face amount of receivable	€1,000,000
Maturity, days	90
Bank prime rate	8.0%
Spread over prime rate on credit line	2.5%
Bank interest (prime + spread), per annum	= 8% + 2.5% = 10.5%
Compensating balance requirement for bank credit line	20%
Export credit insurance fee	1.0%

SOLVE

a. What are the annualized percentage all-in-costs of each alternative?

- **Note:** A compensating balance is a minimum deposit that must be maintained in a bank account by a borrower.

Alternative 1: Bank Credit Line	
Face amount of receivable	€1,000,000
Less: Bank interest expense on receivable	= -€1,000,000 x (90/360) x 10.5% = -€26,250
Less: Compensating balance requirement	= -(€1,000,000 x 20%) = (€200,000)
Net proceeds	= €1,000,000 - €26,250 - €200,000 = €773,750
Annualized all-in-cost of alternative 1	= $\frac{€26,250}{€773,750} \times (360/90) = 0.1357 = 13.57\%$

Alternative 2: Bank Credit Line + Export Credit Insurance	
Face amount of receivable	€1,000,000
Less: Credit insurance fee	= -€1,000,000 x 1.0% = -€10,000
Less: Bank interest expense on receivable	= -€1,000,000 x (90/360) x 5% = -€12,500
Less: Compensating balance requirement	= -(€1,000,000 x 20%) = -€200,000
Net proceeds	= €1,000,000 - €10,000 - €26,250 - €200,000 = €763,750
Annualized all-in-cost of alternative 1	= $\frac{€10,000 + €12,500}{€763,750} \times (360/90) = 0.1178 = 11.78\%$

- **NOTE:** The reason the compensating balance is deducted from net proceeds is that BollyIndia Enterprise does not get that cash and does not earn interest on it.

Problem 12: Swishing Shoe Company. (Cần ôn lại, được 90%)

Assumptions	Values
Face value of the shipment	£400,000
Southampton Footware WACC (per annum)	18.0%
Letter of credit specification on payment after acceptance (days)	120
Banker's acceptance discount rate, 120-day maturity (per annum)	12.0%
Commission for sale of bankers acceptance	2.0%

SOLVE

a. Would Swishing Shoe Company gain by holding the acceptance to maturity, as compared to discounting the bankers' acceptance at once?

Alternative 1	
If Southampton Footware holds the draft for 120 days after the bank has accepted it, Swishing Shoe will receive the face amount of £400,000. The present value of £400,000 received 120 days hence, discounted at Swishing's WACC of 18% per annum (6% for 120 days) is £377,358.49	
Face amount of invoice	£400,000
Discount rate at WACC for 120 days of a 360 day year	$= \frac{1}{1 + 18\% \times 120/360} = 0.9434$
Present value of invoice = Face amount x Discount rate	$= £400,000 \times 0.9434$ $= £377,358.49$

Alternative 2	
Swishing Shoes can sell the banker's acceptance in today's London money market at a 12% per annum discount.	
Face amount of invoice	£400,000
Less: Commission = Commission rate x Face amount	$= -£400,000 \times 2.0\%$ $= -£8,000$
Less: Banker's acceptance discount rate for period (120 days)	$= -£400,000 \times 12\% \times (120/360)$ $= -£16,000$
Amount received now	$= £400,000 - £8,000 - £16,000$ $= £376,000$

=> **Difference between alternatives** = £377,358.49 - £376,000 = £1,358.49

b. Does Swishing Shoe Company incur any other risks in this transaction?

Analysis
Point 1: Swishing's gain should be calculated in present value terms. Swishing will receive £376,000 today by discounting the banker's acceptance. The present value of the £400,000 to be received in 120 days, discounted at Swishing's WACC of 18%, is £377,358.49. The difference is £1,358.49. Swishing would gain, in terms of present value cash, £1,358.49 by waiting 120 days to receive the face amount of the acceptance.
Point 2: In this transaction Swishing has assumed the foreign exchange transaction risk; that is, the risk that the pounds sterling to be received from the export will be worth fewer dollars when received. In part this risk is a function of the time that Swisher must wait to exchange the pounds sterling for dollars. If Swishing discounts the banker's acceptance at the time of sale, it receives dollars at once at the exchange rate then in effect. If Swishing waits 120 days to receive the pounds sterling, Swisher assumes the added risk that the exchange rate will deteriorate between the time of sale and the time of collection. (Of course, Swishing might gain if the pound would buy more, rather than less, dollars in 120 days.)

